

ADVCHAIN: ADVERSARIAL CHAIN-OF-THOUGHT TUNING FOR ROBUST SAFETY ALIGNMENT OF LARGE REASONING MODELS

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ABSTRACT

Large Reasoning Models (LRMs) have demonstrated remarkable capabilities in complex problem-solving through Chain-of-Thought (CoT) reasoning. However, the multi-step nature of CoT introduces new safety challenges that extend beyond conventional language model alignment. We identify a failure mode in current safety CoT tuning methods: the *snowball effect*, where minor reasoning deviations progressively amplify throughout the thought process, leading to either harmful compliance or excessive refusal. This effect stems from models being trained to imitate perfect reasoning scripts without learning to self-correct. To address this limitation, we propose AdvChain, an alignment paradigm that teaches models dynamic self-correction through adversarial CoT tuning. Our method involves constructing a dataset containing Temptation-Correction and Hesitation-Correction samples, where models learn to recover from harmful reasoning drifts and unnecessary cautions. Extensive experiments show that AdvChain significantly enhances robustness against jailbreak attacks and CoT hijacking while substantially reducing over-refusal on benign prompts, achieving a superior safety-utility balance without compromising reasoning capabilities. Our work establishes a new direction for building more robust and reliable reasoning models.

1 INTRODUCTION

Large Reasoning Models (LRMs), which excel at complex problem-solving through explicit Chain-of-Thought (CoT) reasoning, represent a significant advance in artificial intelligence (Guo et al., 2025; Yang et al., 2025; Team, 2025; OpenAI, 2024). By generating a sequence of intermediate reasoning steps before producing a final answer, these models achieve remarkable performance on tasks requiring logic, planning, and explanation (Plaat et al., 2024; Xu et al., 2025a; Chen et al., 2025). However, the multi-step nature of CoT reasoning also introduces a new attack surface, where a single flawed intermediate step can derail an otherwise safe process and corrupt the final outcome, presenting unique and critical challenges for the safety alignment of LRMs (Kuo et al., 2025).

The prevailing paradigm for achieving this, known as Safety CoT Tuning, involves fine-tuning models on curated refusal demonstrations (Jiang et al.; Wang et al., 2025b). In this paradigm, models learn to imitate idealized reasoning chains that safely analyze and reject harmful requests. While effective at eliciting correct refusals on standard benchmarks, we demonstrate that this approach inadvertently instills a critical vulnerability. We term this failure mode the “**Snowball Effect**” in CoT alignment: a small, initial deviation in a reasoning step progressively amplifies throughout the chain, leading to catastrophic outcomes as models cannot self-correct. This effect manifests in two detrimental forms: a **snowballing escalation of harmfulness** for malicious prompts, where reasoning drifts from safe analysis to harmful compliance, and a **snowballing escalation of over-refusal** for benign prompts, where misplaced caution derails helpful intent.

We empirically validate this effect through stepwise evaluation of reasoning chains. For harmful prompts, models often begin with safe analysis but are unable to prevent a gradual descent into

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generating unsafe content. Conversely, for ambiguous but benign prompts, models initially engage constructively but are often trapped by escalating self-doubt, resulting in unnecessary refusals. These dual phenomena reveal that current alignment methods, by teaching models to merely imitate flawless scripts, fail to equip them with the essential capability of dynamic self-correction.

To address this limitation, we propose a new alignment paradigm – adversarial CoT tuning, named AdvChain. Instead of training on exclusively perfect reasoning paths, AdvChain explicitly teaches models to recognize and recover from their own flawed reasoning. Our approach is “adversarial” because it involves fine-tuning models on a novel dataset containing intentionally flawed CoT trajectories that are subsequently corrected. This dataset comprises two key types of self-correcting samples: **Temptation-Correction** samples, which teach the model to halt and reverse a drift towards harmful compliance, and **Hesitation-Correction** samples, which teach it to overcome unnecessary caution and continue providing helpful responses. By training the model with these samples, we aim to break the cognitive inertia that allows the snowball effect to grow unchecked. Extensive experiments demonstrate that AdvChain effectively counteracts the snowball effect. Models tuned with our method show significantly enhanced robustness against both harmful requests and sophisticated CoT hijacking, while simultaneously reducing over-refusal on benign prompts. Furthermore, AdvChain achieves these gains with high data efficiency, comparable with the performance of models trained on $15\times$ more data, without compromising core reasoning capabilities.

Our main contributions are as follows: (1) We identify and empirically validate the “Snowball Effect” in current CoT alignment, characterizing its dual manifestations of escalating harmfulness and over-refusal. (2) We propose adversarial CoT tuning (AdvChain) to train LRMs actively recover from flawed reasoning steps. (3) We construct an adversarial safety reasoning dataset featuring temptation-correction and hesitation-correction samples. (4) Extensive evaluation demonstrate that AdvChain is more robust against attacks and less prone to over-refusal.

2 BACKGROUND AND PRELIMINARIES

2.1 LARGE REASONING MODELS AND CHAIN-OF-THOUGHT

Large Reasoning Models (LRMs) represent an evolution of Large Language Models (LLMs), specifically optimized for complex, multi-step problem-solving (Chen et al., 2025; Xu et al., 2025a; Patil & Jadon, 2025; Liu et al., 2025; Li et al., 2025). Unlike models that produce immediate answers, LRMs excel by generating a sequence of intermediate reasoning steps, a process known as Chain of Thought (CoT) (Wei et al., 2022), before arriving at a final conclusion. This explicit reasoning process, analogous to human cognition, significantly enhances a model’s performance on tasks requiring logical deduction, planning, and mathematical reasoning, as demonstrated by models like DeepSeek-R1 (Guo et al., 2025), Qwen3 (Yang et al., 2025), QwQ (Team, 2025), o1 series OpenAI (2024). Formally, given a user prompt x , an LRM M with parameters θ generates an output $y = M_\theta(x)$. This output can be decomposed into a tuple $y = (c, a)$, where $c = (c_1, c_2, \dots, c_n)$ is the reasoning chain, representing the sequence of intermediate thought steps, and a is the final answer derived from this reasoning chain. While CoT provides valuable transparency into the model’s reasoning process, it also introduces new attack surface that require specialized alignment approaches (Xu et al., 2025b; Zhou et al., 2025a; Zheng et al., 2025; Arrieta et al., 2025; Zhu et al., 2025).

2.2 SAFETY ALIGNMENT OF LARGE REASONING MODELS

The primary goal of safety alignment is to ensure that a model’s outputs adhere to a predefined set of safety principles (e.g., avoiding the generation of harmful, unethical, or biased content) across a wide variety of harmful inputs, denote as $\mathcal{X}_{\text{harm}}$ (Wang et al., 2023; Ma et al., 2025). A prominent method for aligning LRMs is safety CoT tuning (Wang et al., 2025a). Recent approaches include STAR-1 (Wang et al., 2025b) which uses policy-grounded reasoning samples, RealSafe-R1 (Zhang et al., 2025a) with 15k safety-aware trajectories, SafeChain (Jiang et al.) featuring CoT-style safety training, and UnsafeChain (Tomar et al., 2025) focusing on hard case reasoning. These methods fine-tune models on curated datasets $\mathcal{D}_{\text{align}}$ containing safety demonstrations. Each example comprises a tuple $(x_{\text{harm}}, c_{\text{safe}}, a_{\text{safe}})$, where x_{harm} is a harmful prompt, c_{safe} is a safe reasoning chain identifying risks and justifying refusal, and a_{safe} is the final safe response. By learning to imitate these idealized

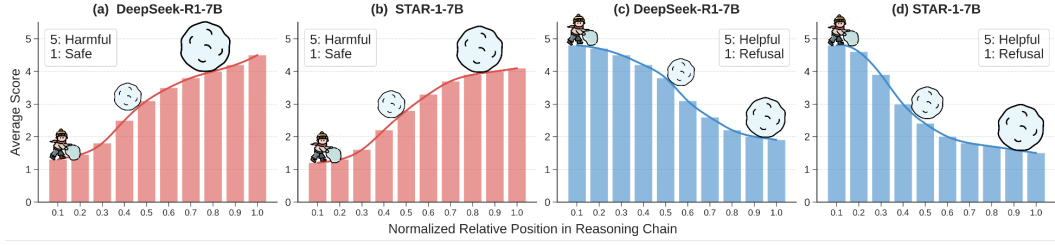


Figure 1: Empirical validation of the Snowball Effect in CoT alignment of current LRMs. (a)-(b): Snowballing escalation of harmfulness. (c)-(d): Snowballing escalation of over-refusal.

reasoning patterns, the model is expected to internalize the underlying safety principles (Zhang et al., 2025b; Zhou et al., 2025b; Wang et al., 2025a).

The model learns through standard language modeling objectives to internalize safety constraints. However, as we demonstrate in subsequent sections, these approaches have fundamental limitations that necessitate more robust alignment paradigms.

This paradigm involves fine-tuning the large reasoning models on a curated dataset, $\mathcal{D}_{\text{align}}$, composed of high-quality safety demonstrations. Each example in this dataset is a tuple $(x_{\text{harm}}, c_{\text{safe}}, a_{\text{safe}})$, where x_{harm} is a potentially harmful user prompt, c_{safe} is a model-generated safe reasoning chain that identifies risks and justifies a refusal according to safety policies, and a_{safe} is the final, succinct, and safe response, which is typically a refusal. The model is then fine-tuned on this dataset by maximizing the standard language modeling objective. By learning to imitate these safety-oriented reasoning patterns, the model is expected to internalize safety alignment capabilities.

3 THE SNOWBALL EFFECT IN CoT ALIGNMENT

We identify a critical failure mode resulting from current CoT alignment methods, which we term the “*snowball effect*”, which describes a process where a small, initial deviation in an intermediate reasoning step progressively amplifies as the reasoning chain unfolds. It occurs because alignment often fails to equip models with the ability to self-correct minor errors, allowing these mistakes to compound and ultimately corrupt the final output. In this section, we empirically demonstrate that this effect manifests in two primary, detrimental forms: a snowballing escalation of harmfulness for harmful prompts and a snowballing escalation of over-refusal for benign prompts.

3.1 SNOWBALLING ESCALATION OF HARMFULNESS

First, our analysis uncovers the critical manifestation of the snowball effect, which we term **snowballing escalation of harmfulness**. This describes a process where the model initiates a safe and valid reasoning path, but a minor deviation in an intermediate step acts as a seed for the snowball. Once flawed step occurs, it begins to gather momentum, progressively corrupting subsequent reasoning and amplifying the initial error into a fully harmful conclusion and output.

Stepwise Safety Analysis. We quantitatively analyze this phenomenon through a stepwise evaluation of reasoning chains generated by DeepSeek-R1-7B and its safety-aligned counterpart STAR-1-7B (Wang et al., 2025b) on harmful prompts from WildJailbreak benchmark. Each reasoning chain is decomposed into individual steps using rule-based newline separation (`\n\n`), and each step receives an independent assessment by GPT-4o on a 5-point safety scale (1 = completely safe, 5 = clearly harmful). We specifically isolate cases where the initial reasoning steps are rated as safe (score ≤ 2) but the final answer is judged harmful by LlamaGuard3 (Llama Team, 2024). For comparative analysis across varying reasoning lengths, the position of each step is normalized to a relative scale from 0.1 to 1.0. This allows us to track the evolution of safety scores and identify the escalation pattern.

Findings. The results, illustrated in Figure 1 (a)-(b), provide empirical evidence for the snowballing escalation of harmfulness. The process does not begin as overtly harmful. Actually, the initial reasoning steps maintain a low score, often averaging below 1.5, as the model correctly identifies the user’s query and initiates a seemingly legitimate analysis. This represents the small, seeding snow-